Database Relationship Analysis Using Joins

Aim:

To analyse relationships between customer, product, order, and payment tables in the EcoCart e-commerce database using SQL JOIN operations and generate useful business reports.

Objectives:

- To combine related tables using SQL JOIN operations.
- To retrieve customer details along with their orders.
- To implement INNER JOIN, LEFT JOIN, and RIGHT JOIN.
- To identify customers who have not purchased products.
- To retrieve complete order details with payment information.
- To analyse customer purchase history.
- To generate customer order, sales, payment, and purchase analysis reports.

Introduction:

The EcoCart e-commerce database stores customer, product, order, payment, and inventory information in separate tables. To retrieve meaningful information from these related tables, SQL JOIN operations are used.

A JOIN combines records from two or more tables based on a related column. It helps analyse customer purchases, order details, payment status, sales performance, and customer spending.

SQL Queries and Implementation

1. INNER JOIN:

INNER JOIN returns only the records that have matching values in both tables.

1. Display customer details along with their orders:

```
SQL> SELECT c.customer_id, c.customer_name, c.email, o.order_id, o.order_date, o.total_amount, o.order_status FROM customer c INNER JOIN orders o ON c.custo
mer_id = o.customer_id;

CUSTOME CUSTOMER_NAME        EMAIL                       ORDER_ID   ORDER_DATE   TOTAL_AMOUNT ORDER_STATUS
------- -------------------- --------------------------- ---------- ------------ ------------ ------------
CUS001  Sanjay Kumar         sanjay@gmail.com            ORD001     01-SEP-26          520.00 Delivered
CUS002  Harini Devi          harini@gmail.com            ORD002     02-SEP-26          250.00 Delivered
CUS003  Praveen Raj          praveen@gmail.com           ORD003     03-SEP-26          600.00 Shipped
CUS004  Keerthana S          keerthana@gmail.com         ORD004     04-SEP-26          400.00 Delivered
CUS006  Pavithra M           pavithra@gmail.com          ORD006     06-SEP-26          850.00 Shipped
CUS007  Ashwin Anand         ashwin@gmail.com            ORD007     07-SEP-26          420.00 Delivered
CUS010  Lavanya M            lavanya@gmail.com           ORD010     10-SEP-26          520.00 Shipped
CUS009  Surya Prakash        surya@gmail.com             ORD009     09-SEP-26          200.00 Pending
CUS008  Nithya R             nithya@gmail.com            ORD008     08-SEP-26          360.00 Pending
CUS005  Dinesh Kumar         dinesh@gmail.com            ORD005     05-SEP-26          180.00 Pending

10 rows selected.
```

2. Display order details with payment information:

```
SQL> SELECT o.order_id, o.order_date, o.total_amount, o.order_status, p.payment_id, p.payment_mode, p.payment_status, p.transaction_amount FROM orders o INN
ER JOIN payment p ON o.order_id = p.order_id;

ORDER_ID   ORDER_DATE   TOTAL_AMOUNT ORDER_STATUS PAYMENT_ID PAYMENT_MODE    PAYMENT_STATUS  TRANSACTION_AMOUNT
---------- ------------ ------------ ------------ ---------- --------------- --------------- ------------------
ORD001     01-SEP-26          520.00 Delivered    PAY001     UPI             Success                        520
ORD002     02-SEP-26          250.00 Delivered    PAY002     Card            Success                        250
ORD003     03-SEP-26          600.00 Shipped      PAY003     Net Banking     Success                        600
ORD004     04-SEP-26          400.00 Delivered    PAY004     UPI             Success                        400
ORD005     05-SEP-26          180.00 Pending      PAY005     Cash            Pending                        180
ORD006     06-SEP-26          850.00 Shipped      PAY006     Card            Success                        850
ORD007     07-SEP-26          420.00 Delivered    PAY007     UPI             Success                        420
ORD008     08-SEP-26          360.00 Pending      PAY008     Card            Pending                        360
ORD009     09-SEP-26          200.00 Pending      PAY009     Net Banking     Failed                         200
ORD010     10-SEP-26          520.00 Shipped      PAY010     UPI             Success                        320

10 rows selected.
```

3. Retrieve products purchased by customers:

```
SQL> SELECT c.customer_name, o.order_id, p.product_name, od.quantity, od.price FROM customer c INNER JOIN orders o ON c.customer_id = o.customer_id INNER JO
IN order_details od ON o.order_id = od.order_id INNER JOIN product p ON od.product_id = p.product_id;

CUSTOMER_NAME        ORDER_ID   PRODUCT_NAME          QUANTITY      PRICE
-------------------- ---------- -------------------- -------- ----------
Sanjay Kumar         ORD001     Organic Oats                1        200
Lavanya M            ORD010     Natural Honey               1        320
Sanjay Kumar         ORD001     Natural Honey               1        320
Harini Devi          ORD002     Aloe Face Cream             1        250
Keerthana S          ORD004     Bamboo Tray                 1        400
Praveen Raj          ORD003     Steel Lunch Box             1        600
Ashwin Anand         ORD007     Eco Diary                   3        140
Nithya R             ORD008     Cotton Tote Bag             2        180
Dinesh Kumar         ORD005     Cotton Tote Bag             1        180
Pavithra M           ORD006     Eco Cotton Kurta            1        850

10 rows selected.
```

2. LEFT JOIN:

LEFT JOIN returns all records from the left table and the matching records from the right table. If there is no match, the right-table columns contain NULL values.

1. Display all customers, including customers who have not placed orders:

```
SQL> SELECT c.customer_id, c.customer_name, o.order_id, o.order_date, o.total_amount FROM customer c LEFT JOIN orders o ON c.customer_id = o.customer_id;

CUSTOME CUSTOMER_NAME        ORDER_ID   ORDER_DATE   TOTAL_AMOUNT
------- -------------------- ---------- ------------ ------------
CUS001  Sanjay Kumar         ORD001     01-SEP-26          520.00
CUS002  Harini Devi          ORD002     02-SEP-26          250.00
CUS003  Praveen Raj          ORD003     03-SEP-26          600.00
CUS004  Keerthana S          ORD004     04-SEP-26          400.00
CUS006  Pavithra M           ORD006     06-SEP-26          850.00
CUS007  Ashwin Anand         ORD007     07-SEP-26          420.00
CUS010  Lavanya M            ORD010     10-SEP-26          520.00
CUS009  Surya Prakash        ORD009     09-SEP-26          200.00
CUS008  Nithya R             ORD008     08-SEP-26          360.00
CUS005  Dinesh Kumar         ORD005     05-SEP-26          180.00

10 rows selected.
```

2. Find customers without any purchases:

```
SQL> SELECT c.customer_id, c.customer_name FROM customer c WHERE NOT EXISTS (SELECT 1 FROM orders o JOIN order_details od ON o.order_id = od.order_id WHERE
o.customer_id = c.customer_id);

CUSTOME CUSTOMER_NAME
------- --------------------
CUS009  Surya Prakash
```

3. Display all products, including products with no sales:

```
SQL> SELECT p.product_id, p.product_name, od.order_id, od.quantity, od.price FROM product p LEFT JOIN order_details od ON p.product_id = od.product_id;

PRODUCT_ID PRODUCT_NAME          ORDER_ID   QUANTITY      PRICE
---------- -------------------- ---------- -------- ----------
PRO001     Organic Oats         ORD001            1        200
PRO002     Natural Honey        ORD001            1        320
PRO003     Aloe Face Cream      ORD002            1        250
PRO006     Steel Lunch Box      ORD003            1        600
PRO005     Bamboo Tray          ORD004            1        400
PRO008     Cotton Tote Bag      ORD005            1        180
PRO009     Eco Cotton Kurta     ORD006            1        850
PRO007     Eco Diary            ORD007            3        140
PRO008     Cotton Tote Bag      ORD008            2        180
PRO002     Natural Honey        ORD010            1        320
PRO004     Bamboo Comb
PRO010     Organic Seed Pack

12 rows selected.
```

3. RIGHT JOIN:

RIGHT JOIN returns all records from the right table and matching records from the left table. If no matching record exists in the left table, its columns contain NULL values.

1.Display all orders with customer information:

```
SQL> SELECT o.order_id, o.order_date, o.total_amount, o.order_status, c.customer_id, c.customer_name FROM customer c RIGHT JOIN orders o ON c.customer_id =
o.customer_id;

ORDER_ID   ORDER_DATE   TOTAL_AMOUNT ORDER_STATUS CUSTOME CUSTOMER_NAME
---------- ------------ ------------ ------------ ------- --------------------
ORD001     01-SEP-26          520.00 Delivered    CUS001  Sanjay Kumar
ORD002     02-SEP-26          250.00 Delivered    CUS002  Harini Devi
ORD003     03-SEP-26          600.00 Shipped      CUS003  Praveen Raj
ORD004     04-SEP-26          400.00 Delivered    CUS004  Keerthana S
ORD005     05-SEP-26          180.00 Pending      CUS005  Dinesh Kumar
ORD006     06-SEP-26          850.00 Shipped      CUS006  Pavithra M
ORD007     07-SEP-26          420.00 Delivered    CUS007  Ashwin Anand
ORD008     08-SEP-26          360.00 Pending      CUS008  Nithya R
ORD009     09-SEP-26          200.00 Pending      CUS009  Surya Prakash
ORD010     10-SEP-26          520.00 Shipped      CUS010  Lavanya M

10 rows selected.
```

2.Find orders where customer details are missing:

```
SQL> SELECT o.order_id, o.customer_id, o.order_date FROM orders o LEFT JOIN customer c ON o.customer_id = c.customer_id WHERE c.customer_id IS NULL;

no rows selected
```

3.Display all payment records with order details:

```
SQL> SELECT p.payment_id, p.order_id, p.payment_mode, p.payment_status, p.transaction_amount, o.order_date, o.total_amount FROM orders o RIGHT JOIN payment
p ON o.order_id = p.order_id;

PAYMENT_ID ORDER_ID   PAYMENT_MODE    PAYMENT_STATUS  TRANSACTION_AMOUNT ORDER_DATE   TOTAL_AMOUNT
---------- ---------- --------------- --------------- ------------------ ------------ ------------
PAY001     ORD001     UPI             Success                       520 01-SEP-26          520.00
PAY002     ORD002     Card            Success                       250 02-SEP-26          250.00
PAY003     ORD003     Net Banking     Success                       600 03-SEP-26          600.00
PAY004     ORD004     UPI             Success                       400 04-SEP-26          400.00
PAY006     ORD006     Card            Success                       850 06-SEP-26          850.00
PAY007     ORD007     UPI             Success                       420 07-SEP-26          420.00
PAY010     ORD010     UPI             Success                       320 10-SEP-26          520.00
PAY009     ORD009     Net Banking     Failed                        200 09-SEP-26          200.00
PAY008     ORD008     Card            Pending                       360 08-SEP-26          360.00
PAY005     ORD005     Cash            Pending                       180 05-SEP-26          180.00

10 rows selected.
```

4.Complete Order Details:

This report combines customer name, product name, quantity, order date, total amount, and payment status.

```
SQL> SELECT c.customer_name, p.product_name, od.quantity, o.order_date, o.total_amount, pay.payment_status FROM customer c JOIN orders o ON c.customer_id =
o.customer_id JOIN order_details od ON o.order_id = od.order_id JOIN product p ON od.product_id = p.product_id LEFT JOIN payment pay ON o.order_id = pay.ord
er_id;

CUSTOMER_NAME        PRODUCT_NAME          QUANTITY ORDER_DATE   TOTAL_AMOUNT PAYMENT_STATUS
-------------------- --------------------- -------- ------------ ------------ ---------------
Sanjay Kumar         Natural Honey                1 01-SEP-26          520.00 Success
Sanjay Kumar         Organic Oats                 1 01-SEP-26          520.00 Success
Harini Devi          Aloe Face Cream              1 02-SEP-26          250.00 Success
Praveen Raj          Steel Lunch Box              1 03-SEP-26          600.00 Success
Keerthana S          Bamboo Tray                  1 04-SEP-26          400.00 Success
Dinesh Kumar         Cotton Tote Bag              1 05-SEP-26          180.00 Pending
Pavithra M           Eco Cotton Kurta             1 06-SEP-26          850.00 Success
Ashwin Anand         Eco Diary                    3 07-SEP-26          420.00 Success
Nithya R             Cotton Tote Bag              2 08-SEP-26          360.00 Pending
Lavanya M            Natural Honey                1 10-SEP-26          520.00 Success

10 rows selected.
```

5.Customer Purchase History:

1.Display all products purchased by a customer:

```
SQL> SELECT c.customer_name, p.product_name, od.quantity, od.price, o.order_date FROM customer c JOIN orders o ON c.customer_id = o.customer_id JOIN order_d
etails od ON o.order_id = od.order_id JOIN product p ON od.product_id = p.product_id WHERE c.customer_id = 'CUS001';

CUSTOMER_NAME        PRODUCT_NAME          QUANTITY      PRICE ORDER_DATE
-------------------- --------------------- -------- ---------- ------------
Sanjay Kumar         Organic Oats                 1        200 01-SEP-26
Sanjay Kumar         Natural Honey                1        320 01-SEP-26
```

2.Find the total amount spent by each customer:

```
SQL> SELECT c.customer_id, c.customer_name, NVL(SUM(o.total_amount),0) AS total_spent FROM customer c LEFT JOIN orders o ON c.customer_id = o.customer_id GR
OUP BY c.customer_id, c.customer_name;

CUSTOME CUSTOMER_NAME         TOTAL_SPENT
------- -------------------- ------------
CUS010  Lavanya M                  520.00
CUS004  Keerthana S                400.00
CUS008  Nithya R                   360.00
CUS002  Harini Devi                250.00
CUS001  Sanjay Kumar               520.00
CUS003  Praveen Raj                600.00
CUS007  Ashwin Anand               420.00
CUS009  Surya Prakash              200.00
CUS006  Pavithra M                 850.00
CUS005  Dinesh Kumar               180.00

10 rows selected.
```

3.Find the number of orders placed by each customer:

```
SQL> SELECT c.customer_id, c.customer_name, COUNT(o.order_id) AS total_orders FROM customer c LEFT JOIN orders o ON c.customer_id = o.customer_id GROUP BY c
.customer_id, c.customer_name;

CUSTOME CUSTOMER_NAME        TOTAL_ORDERS
------- -------------------- ------------
CUS010  Lavanya M                       1
CUS004  Keerthana S                     1
CUS008  Nithya R                        1
CUS002  Harini Devi                     1
CUS001  Sanjay Kumar                    1
CUS003  Praveen Raj                     1
CUS007  Ashwin Anand                    1
CUS009  Surya Prakash                   1
CUS006  Pavithra M                      1
CUS005  Dinesh Kumar                    1

10 rows selected.
```

4.Display the latest purchase details of customers:

```
SQL> SELECT c.customer_name, o.order_id, o.order_date, o.total_amount, o.order_status FROM customer c JOIN orders o ON c.customer_id = o.customer_id WHERE o
.order_date = (SELECT MAX(o2.order_date) FROM orders o2 WHERE o2.customer_id = c.customer_id);

CUSTOMER_NAME        ORDER_ID   ORDER_DATE   TOTAL_AMOUNT ORDER_STATUS
-------------------- ---------- ------------ ------------ ------------
Sanjay Kumar         ORD001     01-SEP-26          520.00 Delivered
Harini Devi          ORD002     02-SEP-26          250.00 Delivered
Praveen Raj          ORD003     03-SEP-26          600.00 Shipped
Keerthana S          ORD004     04-SEP-26          400.00 Delivered
Dinesh Kumar         ORD005     05-SEP-26          180.00 Pending
Pavithra M           ORD006     06-SEP-26          850.00 Shipped
Ashwin Anand         ORD007     07-SEP-26          420.00 Delivered
Nithya R             ORD008     08-SEP-26          360.00 Pending
Surya Prakash        ORD009     09-SEP-26          200.00 Pending
Lavanya M            ORD010     10-SEP-26          520.00 Shipped

10 rows selected.
```

6. Multi-Table Business Reports:

Report 1: Customer Order Report

This report displays the Customer name, order ID, order date, and order status.

```
SQL> SELECT c.customer_name, o.order_id, o.order_date, o.order_status FROM customer c JOIN orders o ON c.customer_id = o.customer_id;

CUSTOMER_NAME        ORDER_ID   ORDER_DATE   ORDER_STATUS
-------------------- ---------- ------------ ------------
Sanjay Kumar         ORD001     01-SEP-26    Delivered
Harini Devi          ORD002     02-SEP-26    Delivered
Praveen Raj          ORD003     03-SEP-26    Shipped
Keerthana S          ORD004     04-SEP-26    Delivered
Pavithra M           ORD006     06-SEP-26    Shipped
Ashwin Anand         ORD007     07-SEP-26    Delivered
Lavanya M            ORD010     10-SEP-26    Shipped
Surya Prakash        ORD009     09-SEP-26    Pending
Nithya R             ORD008     08-SEP-26    Pending
Dinesh Kumar         ORD005     05-SEP-26    Pending

10 rows selected.
```

Report 2: Sales Report

This report displays each product, the total quantity sold, and the total sales revenue calculated from order-detail prices.

```
SQL> SELECT p.product_name, SUM(od.quantity) AS quantity_sold, SUM(od.quantity * od.price) AS total_revenue FROM product p JOIN order_details od ON p.produc
t_id = od.product_id GROUP BY p.product_id, p.product_name;

PRODUCT_NAME          QUANTITY_SOLD TOTAL_REVENUE
--------------------- ------------- -------------
Cotton Tote Bag                   3        540.00
Bamboo Tray                       1        400.00
Aloe Face Cream                   1        250.00
Steel Lunch Box                   1        600.00
Eco Cotton Kurta                  1        850.00
Organic Oats                      1        200.00
Natural Honey                     2        640.00
Eco Diary                         3        420.00

8 rows selected.
```

Report 3: Payment Analysis Report

This report displays the payment mode, number of transactions, and number of successful payments.

```
SQL> SELECT payment_mode, COUNT(payment_id) AS total_transactions, SUM(CASE WHEN payment_status = 'Success' THEN 1 ELSE 0 END) AS successful_payments FROM p
ayment GROUP BY payment_mode;

PAYMENT_MODE    TOTAL_TRANSACTIONS SUCCESSFUL_PAYMENTS
--------------- ------------------ ------------------
Net Banking                      2                  1
Card                             3                  2
Cash                             1                  0
UPI                              4                  4
```

Report 4: Customer Purchase Analysis

1. Find top purchasing customers based on total spending:

```
SQL> SELECT c.customer_name, SUM(o.total_amount) AS total_spent FROM customer c JOIN orders o ON c.customer_id = o.customer_id GROUP BY c.customer_id, c.cus
tomer_name ORDER BY total_spent DESC;

CUSTOMER_NAME        TOTAL_SPENT
-------------------- -------------
Pavithra M                850.00
Praveen Raj               600.00
Lavanya M                 520.00
Sanjay Kumar              520.00
Ashwin Anand              420.00
Keerthana S               400.00
Nithya R                  360.00
Harini Devi               250.00
Surya Prakash             200.00
Dinesh Kumar              180.00

10 rows selected.
```

2. Find the customer or customers with the maximum number of orders:

```
SQL> SELECT c.customer_name, COUNT(o.order_id) AS total_orders FROM customer c JOIN orders o ON c.customer_id = o.customer_id GROUP BY c.customer_id, c.cust
omer_name HAVING COUNT(o.order_id) = (SELECT MAX(order_count) FROM (SELECT COUNT(*) AS order_count FROM orders GROUP BY customer_id));

CUSTOMER_NAME        TOTAL_ORDERS
-------------------- ------------
Lavanya M                       1
Keerthana S                     1
Nithya R                        1
Harini Devi                     1
Sanjay Kumar                    1
Praveen Raj                     1
Ashwin Anand                    1
Surya Prakash                   1
Pavithra M                      1
Dinesh Kumar                    1

10 rows selected.
```

3. Find the customer or customers with the highest spending:

```
SQL> SELECT c.customer_name, SUM(o.total_amount) AS total_spent FROM customer c JOIN orders o ON c.customer_id = o.customer_id GROUP BY c.customer_id, c.cus
tomer_name HAVING SUM(o.total_amount) = (SELECT MAX(customer_spending) FROM (SELECT SUM(total_amount) AS customer_spending FROM orders GROUP BY customer_id)
);

CUSTOMER_NAME        TOTAL_SPENT
-------------------- -----------
Pavithra M                   850
```